Aryan Shekhawat

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EDUCATION

Guru Gobind Singh Indraprastha University **2023 – 2027**
B.Tech in Artificial Intelligence and Data Science - ***CGPA: 8.8*** *Delhi, India*

TECHNICAL SKILLS

Languages: Python, C/C++, SQL (PostgreSQL, MySQL), JavaScript, HTML/CSS
Generative AI & LLMs: LangChain, LangGraph, Agentic Workflows, RAG, LLM Orchestration, Model Context Protocol (MCP), Function Calling, Prompt Engineering, Computer Vision, Langsmith (LLMOps), NLP
Backend & Databases: FastAPI, Prisma ORM, PostgreSQL (pgvector), MySQL, MongoDB, ChromaDB (Vector Databases)
Tools & DevOps: Docker, N8N, Git, Postman, Linux, RESTful API Design, Webhooks
Core Proficiencies: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Software Engineering

WORK EXPERIENCE

Defence Research & Development Organisation (DRDO) - ML Intern **June 2025 – Aug 2025**
Radar Deployment Optimization | ***Tech:*** *Python, NumPy, Rasterio, Matplotlib, DEM/GIS, Streamlit* *New Delhi, India*

- Engineered a hybrid **Genetic Algorithm (GA)** with a novel **Stochastic Heuristic Initialization** to optimize radar placement across difficult terrains.
- Increased total coverage area by **22%** over complex terrains and accelerated algorithm convergence by **30%** compared to standard GA, improving surveillance effectiveness over complex topography.
- Processed and analyzed **Digital Elevation Model (DEM)** and **GIS data** to model terrain constraints.
- Authored Python scripts to automate the generation of interactive coverage maps, terrain analyses, and radar layouts using **Matplotlib, NumPy, and Rasterio**.

PROJECTS

CodeAtlas | React, TypeScript, Node.js, Groq LLM, Mermaid, PostgreSQL

- Developed a full-stack web app that converts any GitHub repository into an **interactive architecture diagram**, with a streamed walkthrough and **AI chat**, helping developers understand unfamiliar codebases faster.
- Implemented an **LLM-powered pipeline** with JSON-schema validation and real GitHub data for file paths, reducing hallucinated output, and added Auth0 login with saved history.

ResumeForge AI | FastAPI, spaCy, LangChain, LLaMA 3.1, SQLAlchemy, React, Auth0

- Developed a full-stack AI platform that analyzes a resume against any job description, giving an **ATS score**, a rewritten resume, **skill gap analysis**, and a tailored cover letter, helping job seekers get past ATS filters.
- Implemented an **NLP parsing pipeline (spaCy)** and a dynamic four-component scoring algorithm with skill weights derived from job description frequency, plus **LLaMA-based** rewriting that avoids fabricated content.

Aletheia | Python, FastAPI, React, ChromaDB, BM25, LangChain, Groq

- Developed a full-stack **RAG search engine** that answers questions from uploaded documents (PDF, Markdown, HTML), with **verified citations** and a confidence score so users know how far to trust each answer.
- Implemented **hybrid retrieval** combining vector search and BM25 keyword search with Reciprocal Rank Fusion and LLM reranking, plus an independent citation verification pass to reduce hallucinated answers.

RESEARCH WORK

Quantum Advantage in Geospatial Analysis | A Hybrid Quantum–Classical Framework for Flood Susceptibility Modeling

- Developed a hybrid framework using a **12-qubit quantum neural network (QNN)** with angle embedding and variational circuits to model flood susceptibility in the GBM Basin, benchmarked against **Random Forest and XGBoost** on a 350-site flood inventory.
- Achieved **0.924 accuracy** and **0.972 AUROC**, outperforming XGBoost by +2.1% accuracy and +1.9% AUROC, with the QNN converging in ∼40% fewer epochs; the first systematic quantum-classical benchmark for flood susceptibility mapping.

ACTIVITIES AND ACHIEVEMENTS

- Completed four professional courses in **Oracle Road to Certification** and achieved **Level 3 Certification**. 2025
- Amongst the **Top 10 Finalists** and Problem Statement Runner-Up at **DEBUGON Hackathon** ONDC. 2025
- Achieved **Arcade Legend** (Ultimate Milestone, Facilitator Program) in **Google Cloud Arcade 2025** 2025
- Served as **Assistant General Secretary** for Utkarsh'25 ADGIPS. 2024
- Selected as **Creative Co-Lead** of the Creative Department at GDGC-ADGIPS. 2025

CERTIFICATIONS

- Oracle Road to Certification
- Anthropic Certification
- ISRO START 2025 (Space Science)
- Hack&Chill 2.0 Co-Lead
- AWS APAC Solutions Architecture
- AICTE × Edunet AI/ML Intern